Module 4: Understanding processes

This module provides progression from the application-oriented work in Physics in Action. Understanding Processes is organised around different ways of describing and understanding processes of change: motion in space and time, wave motion, quantum behaviour. It provides a sound foundation in the classical physics of mechanics and waves and takes the story further, touching on the quantum probabilistic view.

Physical variables are extended from scalars to quantities that add like vectors.

Either section may be covered first. Some teachers may wish to introduce work on vector addition from **4.2** before embarking on work on combining phasors.

The first section of the module is mainly about superposition phenomena of waves with a brief account of the quantum behaviour of photons. This is a rich field for practical physics and learners will have many opportunities to extend their experimental and analytical skills. In addition, the topics provide a picture of the development of theories and understanding over time (HSW1, HSW2, HSW7). Quantum behaviour is discussed through considering possible photon paths, avoiding the wave/particle dichotomy.

4.1 Waves and quantum behaviour

	Learning outcomes	Additional guidance
(a)	*Describe and explain:*	
(i)	production of standing waves by waves travelling in opposite directions	Including graphical treatment HSW1, 2
(ii)	interference of waves from two slits	
(iii)	refraction of light at a plane boundary in terms of the changes in the speed of light and explanation in terms of the wave model of light	*M0.2*
(iv)	diffraction of waves passing through a narrow aperture	HSW6
(v)	diffraction by a grating	
(vi)	evidence that photons exchange energy in quanta $E = hf$ (for example, one of light-emitting diodes, photoelectric effect and line spectra)	limitations of particle and wave models HSW7, 11 *M3.1, M3.2, M3.3, M3.4*
(vii)	quantum behaviour: quanta have a certain probability of arrival; the probability is obtained by combining amplitude and phase for all possible paths	*M1.3*
(viii)	evidence from electron diffraction that electrons show quantum behaviour.	HSW7, 11

(b) *Make appropriate use of:*

(i) the terms: phase, phasor, amplitude, probability, interference, diffraction, superposition, coherence, path difference, intensity, electronvolt, refractive index, work function, threshold frequency.

(c) *Make calculations and estimates involving:*

(i)	wavelength of standing waves	end corrections not required
(ii)	Snell's Law, $n = \frac{\sin i}{\sin r} = \frac{c_{1\text{st medium}}}{c_{2\text{nd medium}}}$	*M03, M0.6, M4.5*
(iii)	path differences for double slits and diffraction grating, for constructive interference $n\lambda = d\sin\theta$ (both limited to the case of a distant screen)	angles may be given in degrees or radians, the use of the small angle approximation is expected. *M0.6, M4.6, M4.7*
(iv)	the energy carried by photons across the spectrum, $E = hf$	*M0.2, M2.3*
(v)	the wavelength of a particle of momentum p, $\lambda = \frac{h}{p}$	As given by the de Broglie relationship *M0.2, M2.3*

(d) *Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):*

(i)	using an oscilloscope to determine frequencies	links to **4.1a(i)** **PAG5**
(ii)	determining refractive index for a transparent block	links to **4.1c(ii)** **PAG6**
(iii)	superposition experiments using vibrating strings, sound waves, light and microwaves	links to **4.1a(i), b(i), c(i)** **PAG5**
(iv)	determining the wavelength of light with a double-slit and diffraction grating	links to **4.1a(ii), a(v), c(iii)** **PAG5**
(v)	determining the speed of sound in air by formation of stationary waves in a resonance tube	links to **4.1a(i), c(i)** **PAG5**
(vi)	determining the Planck constant using different coloured LEDs.	links to **4.1a(vi), c(iv)** **PAG6**